

Defense Against FGSM Adversarial Attacks

1. Introduction

This report demonstrates the implementation and effectiveness of Adversarial Training as a defense mechanism against Fast Gradient Sign Method (FGSM) attacks on a Convolutional Neural Network (CNN) trained on the CIFAR-10 dataset.

2. Defense Methodology: Adversarial Training

2.1 Concept

Adversarial training is a robust defense technique where the model is trained on both clean and adversarially perturbed examples during the training process. This exposure to adversarial samples during training helps the model learn features that are more resilient to adversarial perturbations.

2.2 FGSM Attack Generation

The Fast Gradient Sign Method generates adversarial examples by:

- Computing the gradient of the loss with respect to the input image
- Taking the sign of the gradient: $\text{sign}(\nabla_x L(\theta, x, y))$
- Adding perturbation: $x_{\text{adv}} = x + \varepsilon \times \text{sign}(\nabla_x L(\theta, x, y))$
- Clamping to valid pixel range $[0, 1]$

Mathematical formulation:

$$x_{\text{adv}} = \text{clip} \left(x + \varepsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)), 0, 1 \right)$$

Where:

- x : Original input image
- ε : Perturbation magnitude (controls attack strength)
- $J(\theta, x, y)$: Loss function
- θ : Model parameters

2.3 Adversarial Training Process

During each training iteration:

- **Clean examples:** Forward pass on original images
 - Compute loss: $L_{\text{clean}} = \text{CrossEntropy}(f(x), y)$
- **Adversarial examples:** Generate FGSM attacks on-the-fly
 - Create: $x_{\text{adv}} = \text{FGSM}(x, y, \varepsilon)$
 - Compute loss: $L_{\text{adv}} = \text{CrossEntropy}(f(x_{\text{adv}}), y)$
- **Combined loss:** Mix both losses with weight α

$$L_{\text{total}} = (1 - \alpha) \times L_{\text{clean}} + \alpha \times L_{\text{adv}}$$

- **Backpropagation:** Update model parameters using combined loss

Key parameters used:

- ε (epsilon): 0.03 ($\approx 8/255$ pixel intensity)
- α (alpha): 0.5 (equal weight to clean and adversarial losses)
- Training epochs: 70
- Optimizer: AdamW with learning rate 0.001

2.4 Model Architecture

A custom CNN with the following structure:

- 3 convolutional blocks with BatchNorm and ReLU
 - Block 1: 64 filters
 - Block 2: 128 filters
 - Block 3: 256 filters
- MaxPooling after each block ($32 \rightarrow 16 \rightarrow 8 \rightarrow 4$)
- Fully connected layers: $256 \times 4 \times 4 \rightarrow 512 \rightarrow 10$ classes
- Regularization: Dropout (0.3), Label Smoothing (0.1)

3. Experimental Results

3.1 Quantitative Analysis

The evaluation was performed on the CIFAR-10 test set across multiple attack strengths (ϵ values):

Epsilon	Normal Model	Defense Model	Improvement
0.000	88.50%	85.22%	-3.28%
0.010	49.40%	60.28%	+10.88%
0.020	44.91%	56.67%	+11.76%
0.030	40.85%	53.14%	+12.29%
0.050	34.75%	46.91%	+12.16%
0.070	30.47%	41.85%	+11.38%
0.100	25.35%	35.10%	+9.75%
0.150	20.73%	28.02%	+7.29%
0.200	18.05%	23.23%	+5.18%

3.2 Key Observations

Clean Accuracy ($\epsilon = 0.0$)

- Normal Model: 88.50%
- Defense Model: 85.22%
- Trade-off: -3.28% drop in clean accuracy

The slight decrease in clean accuracy is expected and acceptable, as the model learns to prioritize robustness over marginal gains on clean data.

Adversarial Robustness ($\epsilon = 0.03$)

- Normal Model: 40.85% (47.65% drop from clean)
- Defense Model: 53.14% (32.08% drop from clean)
- Improvement: +12.29 percentage points

At the training epsilon ($\epsilon = 0.03$), the defense model shows substantial improvement, maintaining over 53% accuracy compared to only 41% for the normal model.

Strong Attacks ($\epsilon \geq 0.05$)

Even under stronger attacks:

- At $\epsilon = 0.05$: Defense maintains 46.91% vs 34.75% (normal)
- At $\epsilon = 0.10$: Defense maintains 35.10% vs 25.35% (normal)
- Consistent improvement of $\sim 10\%$ across all attack strengths

3.3 Robustness-Accuracy Trade-off

The results demonstrate the classic robustness-accuracy trade-off:

- Small sacrifice in clean accuracy (-3.28%)
- Significant gain in adversarial robustness (+12.29% at $\varepsilon = 0.03$)
- Generalizes to various attack strengths (consistent improvements)

The trade-off ratio is highly favorable:

$$\text{Robustness Gain} / \text{Clean Accuracy Loss} = 12.29\% / 3.28\% \approx 3.75\times$$

3.4 Attack Success Rate

Calculating attack success rate (ASR):

$$\text{ASR} = \frac{\text{Clean Accuracy} - \text{Adversarial Accuracy}}{\text{Clean Accuracy}} \times 100\%$$

At $\varepsilon = 0.03$:

- Normal Model ASR: $(88.50 - 40.85) / 88.50 = 53.84\%$
- Defense Model ASR: $(85.22 - 53.14) / 85.22 = 37.66\%$

The defense reduces attack success rate by 16.18 percentage points, meaning adversarial training makes the model significantly more resistant to FGSM attacks.

4. Visual Analysis

4.1 Robustness Curve

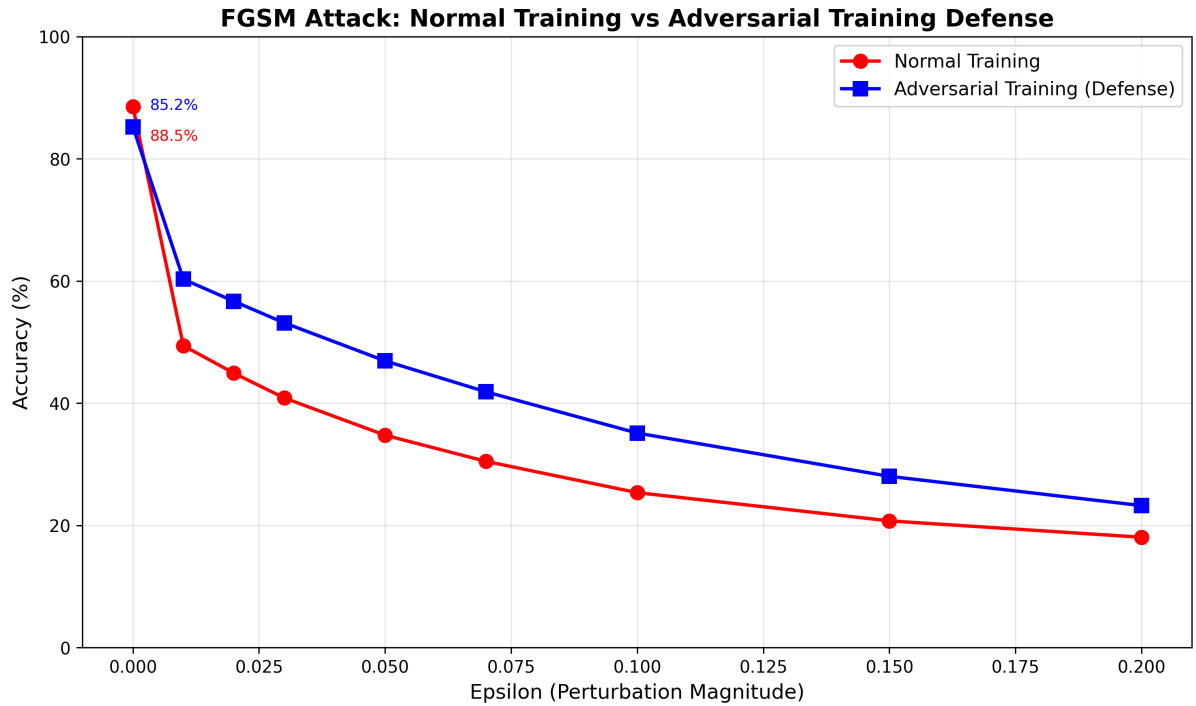


Figure 1:

- Red line (Normal Model): Rapid accuracy degradation as ε increases
- Blue line (Defense Model): More gradual decline, maintaining higher accuracy throughout
- Divergence point: Maximum separation occurs around $\varepsilon = 0.03\text{--}0.05$
- Asymptotic behavior: Both models converge to lower accuracy at very high ε (> 0.15)

4.2 Adversarial Examples Visualization

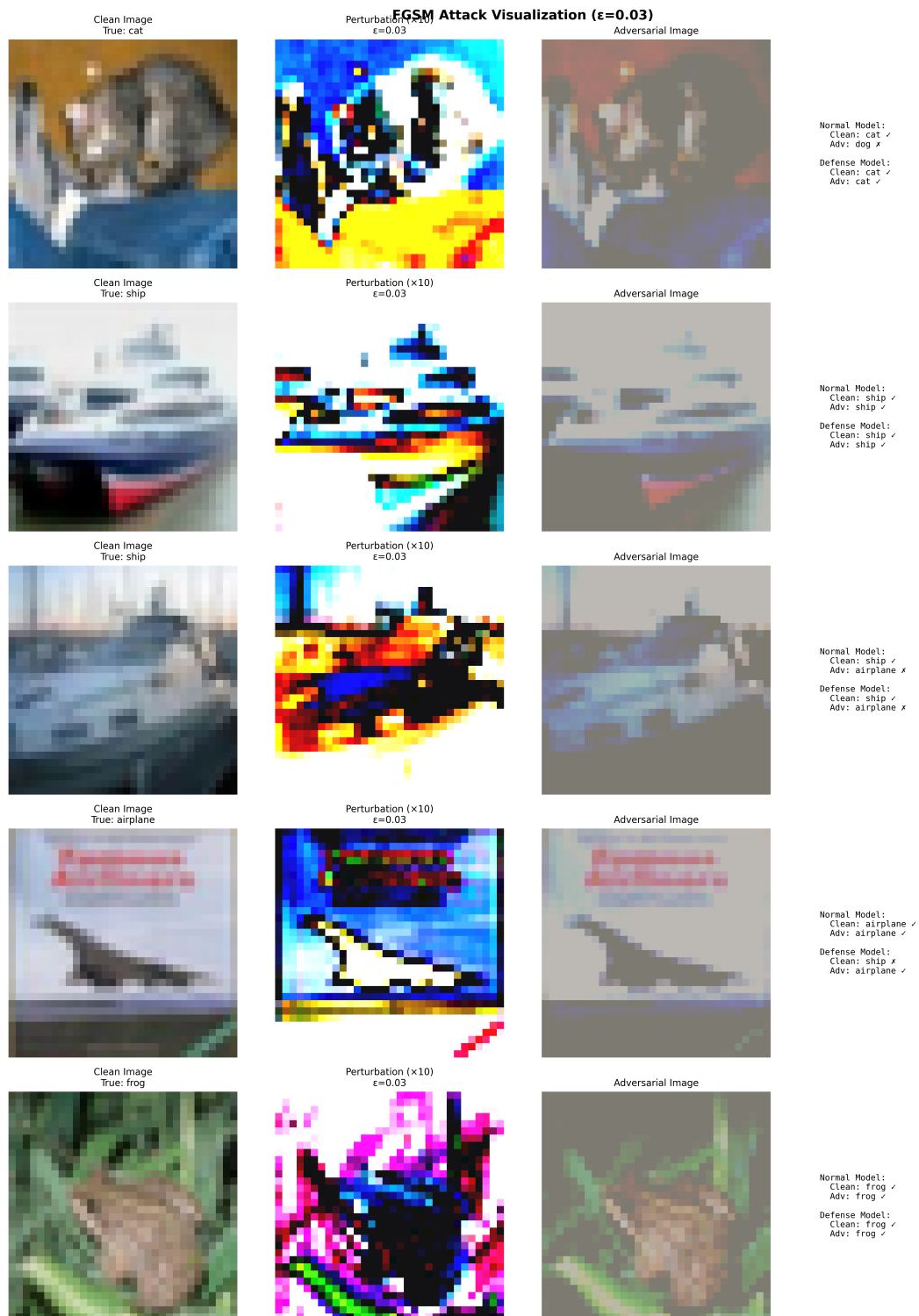


Figure 2:

- Original images: Clean CIFAR-10 test samples
- Perturbations: FGSM noise (amplified 10× for visibility) - appears as subtle noise

patterns

- Adversarial images: Perturbed versions (visually nearly identical to originals)
- Predictions comparison:
 - Normal model: Often misclassifies adversarial examples
 - Defense model: Correctly classifies most adversarial examples

Key insight: The perturbations are imperceptible to humans but significantly affect the normal model, while the defended model remains robust.

5. Defense Effectiveness Analysis

5.1 Strengths

- Significant robustness improvement across all attack strengths
- Minimal clean accuracy degradation (only 3.28%)
- Simple to implement - no architectural changes required
- Generalizes well to different epsilon values
- Training-time defense - no inference overhead

5.2 Limitations

- Computational cost: Training time increases ($\sim 2\times$ due to adversarial example generation)
- Not transferable: Defense is specific to FGSM-style attacks
- Moderate protection: Does not achieve 100% robustness
- Clean accuracy trade-off: Some performance sacrifice on benign data

5.3 Comparison to Baseline

Metric	Normal Training	Adversarial Training	Change
Clean Accuracy	88.50%	85.22%	-3.28%
Robust Accuracy ($\varepsilon = 0.03$)	40.85%	53.14%	+12.29%
Attack Degradation	47.65%	32.08%	-15.57%
Training Time	$\sim 1\times$	$\sim 2\times$	+100%

6. Theoretical Understanding

6.1 Why Adversarial Training Works

Adversarial training works by:

- Smoothing decision boundaries: The model learns smoother, more continuous decision surfaces
- Feature robustness: Forces the model to rely on more robust, meaningful features rather than spurious correlations
- Worst-case optimization: Trains on examples near decision boundaries, improving robustness
- Implicit regularization: Acts as a data augmentation technique that improves generalization

6.2 Mathematical Intuition

Standard training minimizes:

$$\min_{\theta} E_{(x,y) \sim D} [L(\theta, x, y)]$$

Adversarial training solves:

$$\min_{\theta} E_{(x,y) \sim D} \left[\max_{\delta \in S} L(\theta, x + \delta, y) \right]$$

This min-max formulation finds parameters θ that perform well even on worst-case perturbations within a bounded set S .

7. Conclusions

7.1 Summary

This implementation demonstrates that adversarial training is an effective defense against FGSM attacks:

- +12.29% robustness improvement at standard attack strength ($\varepsilon = 0.03$)
- Only -3.28% clean accuracy loss - acceptable trade-off
- Consistent protection across various attack strengths
- Practical implementation without architectural changes

7.2 Practical Recommendations

- Use adversarial training when robustness is critical (security-sensitive applications)
- Balance α parameter based on deployment requirements:
 - Higher $\alpha (> 0.5)$: Prioritize robustness
 - Lower $\alpha (< 0.5)$: Prioritize clean accuracy
- Choose ε based on threat model: Common values are 2/255, 4/255, 8/255
- Monitor both metrics: Track clean and adversarial accuracy during training

7.4 Final Remarks

The results conclusively show that adversarial training successfully defends against FGSM attacks, transforming a vulnerable model (40.85% robust accuracy) into a significantly more robust one (53.14% robust accuracy) with minimal clean accuracy sacrifice. This demonstrates the effectiveness of adversarial training as a practical defense mechanism for real-world deployments where adversarial robustness is required.